

A computational ore microscope: per-pixel Stokes polarimetry for characterisation of UG2 ore

REEFPRIINT (developed as KHANYA) — Team Sonar

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CHALLENGE: COMPUTER VISION FOR REAL-TIME MINERALOGICAL CHARACTERISATION

The problem. PGE recovery from UG2 ore is governed by base-metal sulphide deportment: pentlandite is the principal PGE host and floats, pyrrhotite is depressed and carries little PGE, and fine chromite reporting to concentrate is a smelter penalty. Resolving that assemblage today means SEM-based automated mineralogy — QEMSCAN or MLA — accurate, but offline, slow, capital-intensive, and therefore never in the control loop. The obvious optical shortcut fails on physics rather than on engineering: hyperspectral reflectance identifies minerals by molecular absorption features, and chromite — 50–75 vol% of UG2 — is an opaque spinel with none. On chromitite, a spectrometer is a brightness meter.

Our approach. Opaque ore minerals have been identified since the 1940s by quantitative specular reflectance, bireflectance and anisotropy under crossed polars, and published automated optical mineralogy — the Castroviejo–Pirard line, CAMEVA, AMCO — classifies on multispectral reflectance alone. We add an axis reflectance does not contain: recovery of the **full linear Stokes vector at every pixel** from a rotating-analyser image series. Reflectance cannot reliably separate pentlandite (cubic, isotropic) from pyrrhotite (moderately bireflectant, anisotropic); the per-pixel Stokes signature can — and that separation governs PGE deportment and flotation response. Polarised-light *imaging* in ore microscopy is established prior art (Pirard, Lebichot & Krier, 2007); per-pixel Stokes vector recovery is not, and that narrow claim is the one we defend.

Methods and technologies. Segmentation of polished-section imagery (PyTorch, Apache-2.0 backbones), reflectance calibrated against the IMA/COM Quantitative Data File, harmonic inversion for the Stokes parameters, and conformal prediction for calibrated per-prediction uncertainty. Three decisions distinguish the build. First, **acquisition geometry is measured, not assumed**: a rotating analyser modulates at the second harmonic and yields the Stokes vector; stage rotation under crossed polars modulates at the fourth and yields only extinction depth. Fitting the Stokes model to a stage rotation returns zero anisotropy for every anisotropic grain, silently — so the inversion refuses to run until geometry is confirmed from the data. Second, **every quantity carries its provenance** — measured, cited, stipulated, assumed — through arithmetic, so an assumption cannot be laundered into a reported result. Third, **the system abstains**: at ore transitions it emits a conservative default with a stated reason rather than holding the previous setpoint. It runs fully offline on one laptop, on permissively licensed, SBOM-tracked dependencies.

Evidence to date. The Stokes inversion resolves a 40.4× separation between isotropic and anisotropic response on a synthetic phantom with analytic ground truth — a phantom result, not an ore result. Applied to public reflected-light data (LumenStone S3 v2; 29 sections, 116 000 pixels), our geometry discriminator found neither harmonic above detection threshold — evidence that the published “rotation sequences” are stage rotations, not analyser rotations. We report that negative finding and route to a fourth-harmonic extinction estimator rather than force the convenient inversion. We have also quantified the evaluation trap this domain invites: on synthetic data whose only signal is section identity, patch-level splitting flattens error by a factor of 126 against an honest locality-grouped split. All results run under 249 automated tests.

Expected outcomes and impact. Per-particle modal mineralogy and base-metal sulphide deportment from reflected-light optics, carrying calibrated uncertainty and an explicit refusal mechanism, feeding three operator-facing indices: fine-chromite entrainment risk, naturally-floating-gangue load for depressant dosing, and stockpile oxidation. A ~R5 000 instrument is costed as a design; no hardware is built, and no claim here depends on one.

Falsifiability. We hold a pre-registered null hypothesis: after controlling for Cr₂O₃ and pyroxene fraction, texture carries no additional predictive signal. It is tested by nested model comparison with cluster-robust inference grouped by locality and reported whichever way it resolves; if it cannot be rejected, we will say so publicly and pivot to the oxidation index.

Sources: public data only (LumenStone, IronOreRLM), IMA/COM reflectance reference data, and Craig & Vaughan (open access, MSA). No proprietary Mintek data is used. AI coding assistants supported implementation and drafting; the scientific direction and system design are the team's own, evidenced in a continuous public commit history.